

M2 "FunPhys" - Modeling Project

Exploring transient classification using deep generative model

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Abstract: In this work, we present the results of the binary classification (type Ia or non-Ia) of 1606 supernovae from the ZTF catalogue following the ParSNIP machine learning classification tool, so far adapted to Pan-STARRS supernova surveys. We explain in detail how we adapt the ZTF dataset to the standardised format required by the ParSNIP algorithm to generate predictions. We find that for an algorithm pre-trained from the PS1 dataset, the classification of the Ia supernovae from this ZTF dataset sample is accurate 79% of the time. However, the algorithm wrongly predicts the classification of non-Ia supernovae 55% of the time. These results are compared and discussed to the already known results obtained after the predictions from the PS1 dataset. We also present four supernovae lightcurves to analyse further and discuss the potential reasons for our findings. This work gives an indication of the expected results we might obtain if conducting a complete classification study, including all different supernovae' classifications. It also provides hope for the accuracy of the predictions on future, improved classification studies on ZTF using ParSNIP.

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0 Introduction

Machine Learning (ML) algorithms have become increasingly crucial in astrophysics in recent years. From the precise estimation of black holes masses [Lin et al., 2022] to the simulation of galaxy formation [Buncher and Carrasco Kind, 2020] and the detection of exoplanets [Dattilo et al., 2019], among others, these algorithms are a precious tool to analyse and interpret the vast amounts of data and help physicists gain insights into the nature and evolution of the universe.

ML algorithms, and more precisely neural networks algorithms, are also starting to be used to help classify supernovae. Supernovae are powerful and fascinating astronomical events occurring when a star reaches the end of its life and explodes. Usually, during this process, the luminosity of the star increases significantly, which makes the study of the richness of the phenomenon taking place possible directly from the earth. There are several types of supernovae, each characterised by their occurring physical processes, light curves, and spectral features. The most common types of supernovae are Type Ia supernovae. They occur in binary star systems where one star is a white dwarf, and the other is a red giant or another white dwarf. The white dwarf accretes matter from its companion star until it reaches a certain mass, causing the star to explode after a nuclear fusion reaction. Type Ia supernovae are helpful as standard candles for measuring cosmological distances thanks to their uniform luminosity. Moreover, there exists Type Ib and Ic supernovae. They are similar to Type Ia supernovae, they also result from the explosion of a star's core, but they are thought to be caused by stars alone rather than binary systems. Type II supernovae occur when a massive star reaches the end of its nuclear fusion life and collapses under its gravity. This collapse triggers a powerful explosion that releases energy and produces heavy elements. Type II supernovae also decline in different types as IIb, IIc, II-L, II-P...

The most recently developed algorithm to help the classification of such astronomical objects is called ParSNIP [Boone, 2021b] and uses an alternative version of a variational autoencoder (VAE) [Kingma and Welling, 2013]. The VAE has been shown to be efficient for photometric classifications. Such a neural network is composed of an encoder and a decoder. The first encodes the input data points in lower dimensions in the so-called latent space representation capturing the key features and patterns of the data. The decoder then generates a reconstructed output similar to the input data from the probability distribution functions constructed in the latent space. This algorithm also describes how the spectra of transients objects evolve over time by integrating physics model on how light propagates through the universe before being observed.

The goal of this work is to evaluate the latter ParSNIP classification tool trained from the Pan-STARRS1 Medium-Deep Survey (PS1) [Chambers et al., 2016] to a new dataset of supernovae discovered by the Zwicky Transient Facility (ZTF) [Graham et al., 2019]. This will give us a first insight into whether the ParSNIP algorithm can be well adapted to such a dataset, namely if it correctly predicts the classification of the registered supernovae. Hence we will compare the predictions' results with those already obtained from PS1.

In Section 1, we detail the used ZTF and PS1 datasets, the methods employed to adapt the format to the ParSNIP algorithm, and how we used this algorithm. In Section 2, we present the results of our study. Namely, we introduce the confusion matrices following the classification of the supernovae in the two datasets and discuss a few relevant cases. In section 3, we conclude and present further implementable works and improvements.

1 Data and Methodology

In this work, we generated classification predictions using the ParSNIP classification tool for the two sub-datasets of PS1 and ZTF. The Pan-STARRS1 telescope, located at Haleakala Observatory in Hawaii, US, has some of the most significant volumes of astronomical data. On the other hand, ZTF is composed of an imager on the Palomar, San Diego, US, 48-inch Oschin (type Schmidt) telescope and a spectrograph on the Palomar 60-inch telescope optimised for spectral classification of bright transient explosive objects. It could be more adapted to the study of supernovae, and that is why it is relevant to understand how to apply the algorithm to this new type of data, to which the algorithm is not configured.

First, we reproduced the predictions on a smaller subset of the PS1 dataset. It had already been done for the whole dataset by my supervisor. Still, this step is relevant to properly understand the format of PS1, on which the ParSNIP algorithm is adapted. The PS1 dataset is available in an 'h5' format, a file format used to store and organise large amounts of data in a hierarchical data format (HDF), which is convenient for managing extensive scientific data. In an HDF, data is organised into groups and datasets containing numerical data, text, or other data types. The groups and datasets can be nested to create a hierarchical structure, representing complex relationships between data. It is supported by Python and provides a standardised way to store and share data, making collaboration between researchers practical. For all those reasons, it is convenient to understand this format correctly to further manage the conversion of the ZTF dataset into HDF. To do this, we read the light curves of the supernovae recorded by the PS1 data file using `lcdata.read_hdf5` function from the `lcdata` package, which is a package convenient to deal with vast datasets of astronomical time series, such as the lightcurves of supernovae, and is the dataset type adapted to the ParSNIP algorithm. The variable type is similar to an array, and we thus selected the $N = 1123$ first lightcurves corresponding to the 1123 first recorded supernovae in the PS1 set with a defined redshift value in a similar manner as one would do to choose a portion of elements of an array. Then, we recreated an HDF file using `write_hdf5` function containing the pieces of information of the subset. We used the command introduced in [Boone, 2021a] to generate predictions on this smaller subset. This `parsnip_predict` script is used to create classification forecasts given a `lcdata` lightcurves dataset from the HDF format file and a pre-trained ParSNIP model. My supervisor did the pre-training model of the ParSNIP algorithm on the whole PS1 dataset.

Overall, in this task, we looked on the internet how to deal with HDF files: how to read them, understand their structure, store the supernovae lightcurves in variables and then rewrite in another HDF file format containing all the remaining pieces of information of the considered smaller subset. To generate classification predictions of the created file containing the data PS1 subset, as mentioned, we adapted the command introduced by the ParSNIP creator to our file names. The algorithm from already existing training then generated the predictions.

Secondly, the core of this numerical work was to generate predictions on the ZTF dataset. Since this dataset is accessible in the common `Pandas.DataFrame` from the `Pandas` package, it is necessary to find a way to convert this `Pandas.DataFrame` type into a `lcdata` type, which will then be written into an HDF format file easily and be compatible with the classification algorithm. Following what is indicated on how to use any `lcdata` datasets [Boone, 2021c], for each supernova encountered in the

Pandas dataset, we accessed and stored in different variables information of each studied supernova. Such variables are the redshift and its error, the type, and the full light curve for the various filters. The light curve records the flux of received photons by the detector as a function of time. ZTF uses six photon filters, which are denoted by the letters g, r, i. They cover a range of wavelengths from the blue end of the visible spectrum (g filter) to the near-red (i filter). Each filter captures specific colours of light and blocks out the unwanted wavelengths. The g and r filters capture blue and green light, respectively, while the i captures the red light. The g and r filters being optimised for detecting supernovae are the ones of our interest. We made sure to rename those filters to adapt to the string format used by the ParSNIP algorithm. Each filter is indeed written differently in the ZTF dataset, and the one adapted to the ML algorithm.

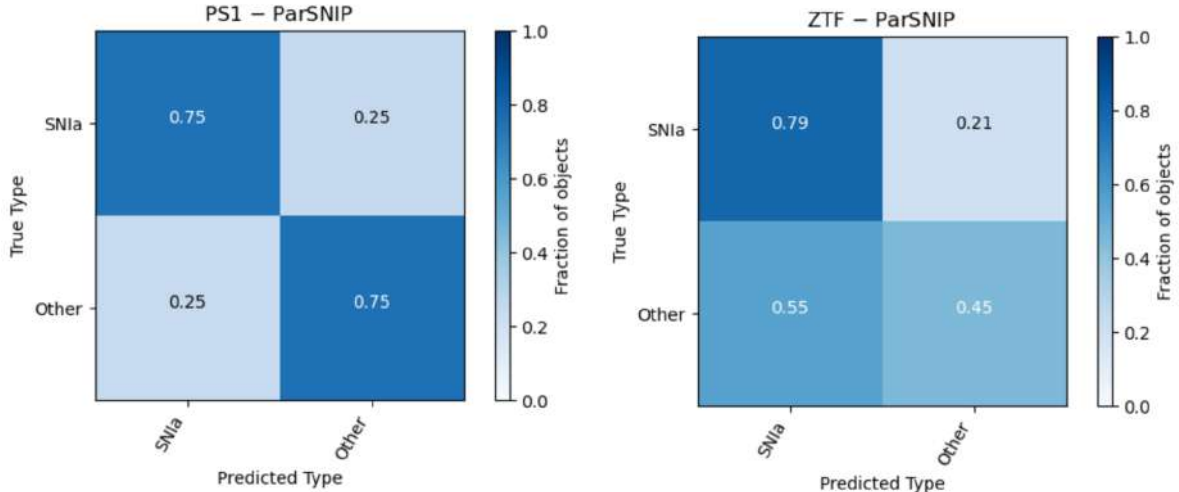
lcdm datasets contains one `astropy.Table.Table` dataset, which gathers the information on the light curves of the supernovae and a 'metadata' environment gathering the basic information around the studied supernova. Hence, we then created an `astropy.Table.Table` dataset gathers the flux information, the corresponding time series, the error on the flux and the band filters in this table. We also created a 'metadata' environment, required by any `lcdm` dataset, gathering the supernova's redshift, type or name.

Because the ParSNIP algorithm does not read the classification of the supernovae in the same way as it is registered in ZTF, we had to adapt the name of their type. Since ZTF has extra classifications detail (ZTF makes the difference between different Ia-type as well, for instance), we ignore those more detailed classifications as a first application and hence adjusted any more classified Type Ia supernova to the general Type Ia and all the others (non-Ia) to "Unknown" (for simplicity here "Unknown" is used to refer to any types different from Ia, because the ParSNIP algorithm accepts either classified objects or "Unknown" objects). We do this to reduce the predictions, to a first extent, to a binary problem, namely predicting the supernova's type to be either Ia or not. Hence, an improvement of this work would be to do it in the general case, classify all the supernovae to conduct a full classification prediction of ZTF.

2 Results and Discussions

In this section we summarise and discuss the results of the predictions made by the ParSNIP model on two different subsets of PS1 and ZTF.

First, we plot the resulting confusion matrices after the generated predictions made by the ParSNIP algorithm for the two PS1 and ZTF datasets. A confusion matrix is a table used to evaluate the performance of an ML model on a set of data where the actual values are known. The matrix summarises the number of correct and incorrect predictions made by the model. It is hence a valuable tool for measuring the performance of ML algorithms. Those matrices for the two datasets are shown in Figure 1. Here, each row shows what fraction of a supernova of a given type is assigned to each label by the classification model. To compare with the ZTF dataset and stay simple in our analysis, we decided to show the confusion matrix resulting from a binary classification: either the supernovae were classified as "SNIa" or "others". We find that the ParSNIP model can accurately classify type Ia supernovae 75% (79%) of the time and non-Ia types 75% (45%) of the time in the case of the PS1 (ZTF) subset. These findings suggest that the ParSNIP algorithm seems to slightly better classify type Ia supernovae on the ZTF subset than on PS1 confirming the relevance on conducting future classifications on ZTF rather



(a) 1123 studied supernovae from PS1

(b) 1606 studied supernovae from ZTF

Figure 1: Confusion matrix for the ParSNIP predictions on smaller subsets of PS1 (a) and ZTF (b) datasets. Each entry shows the fraction of transients of the type labeled on the left side of the plot that are assigned the corresponding type labeled on the bottom of the plot.

than PS1 thanks to the increased accuracy of ZTF lightcurves, as mentioned. However, the algorithm poorly predicted the classification of non-Ia supernovae in the ZTF sample, while it predicted those types in the PS1 sample relatively well. Hence, even though the ParSNIP algorithm looks efficient in predicting type Ia supernovae, it over-predicts the number of type Ia supernovae in the sample.

To push the study further and analyse what could cause the ParSNIP algorithm to badly predict the classification of some of the supernovae in the ZTF sample, we plotted four different light curves of supernovae corresponding to the different scenarios predicted by the confusion matrix: type Ia supernovae well and not well identified by the algorithm, and non-Ia type supernovae well and non-well recognised.

In Figure 2 and 3 we present two such examples, a well identified supernova and a badly predicted one, respectively. As we can see, the well identified type Ia supernova has a lot of data points and fit perfectly the expected light curves (lines) in the two filters. On the other hand, the badly predicted type Ia supernova was lacking crucial data points. First, only the g filter was available for this supernova and, secondly the extracted flux is really poor. These results reinforces the claim that the ParSNIP algorithm predicts well the classifications of type Ia supernovae. Indeed, it is not a surprise that the algorithm fails to predict the classification of supernovae when so little data are available.

In Figure 4 and 5 we present two example of well and badly predicted non-Ia supernovae, respectively. The well identified non-Ia supernova seems in this case to be attributed to the error. The type of this supernova might be unknown, the algorithm predicted well its type thanks to the measurement errors. On the other hand, it predicted the other supernova's type to be Ia while it had been known to be another type. We believe here that this supernova might actually be of type Ia but had been badly identified by the spectrograph during the measurements. Yet, it remains not clear and further studies are required.

Conducting the study also on the PS1 dataset enabled us to compare the accuracy

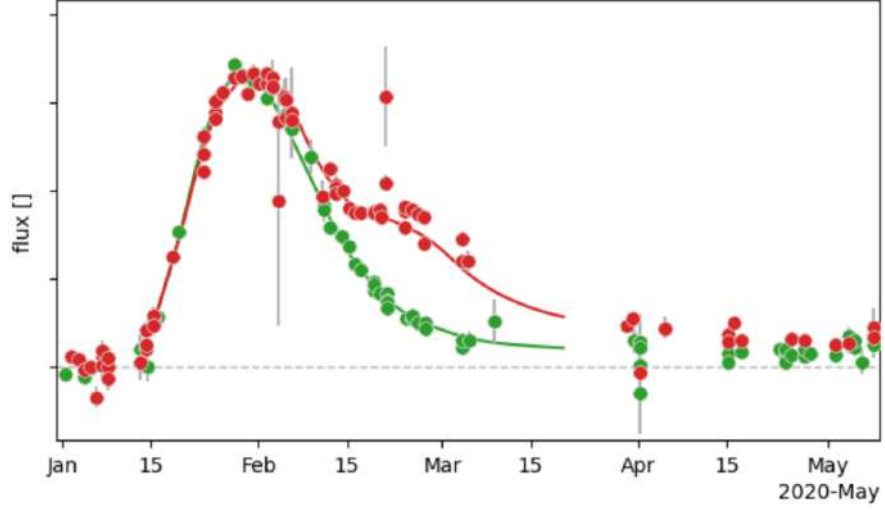


Figure 2: Light curve of type Ia-supernova "ZTF17aadlxmv" at estimated redshift $z = 0.062$. The photons in the two g and r band filters are shown. The points represent the photons data while the lines represent the best physical light curve fitting.

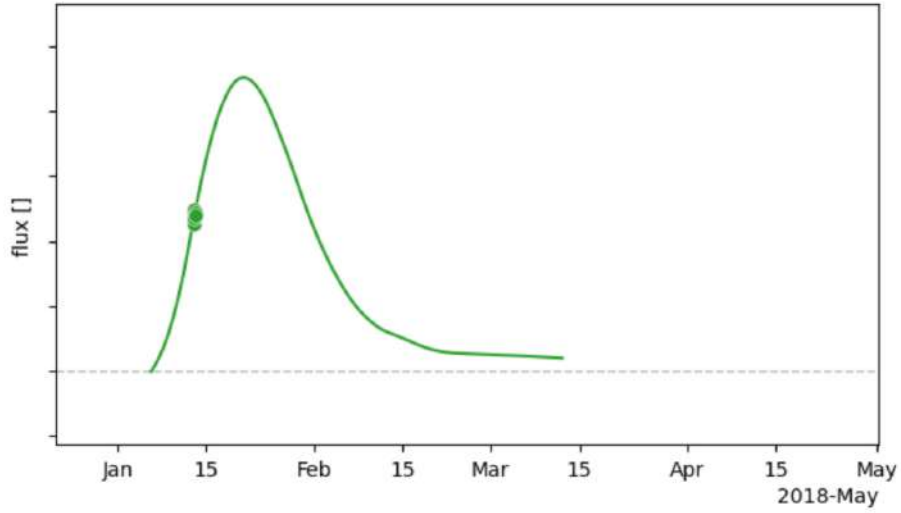


Figure 3: Light curve of type Ia-supernova "ZTF18aaadqua" at estimated redshift $z = 0.083$. Only the photons in two g band filters were recorded. The points represent the photons data while the lines represent the best physical light curve fitting.

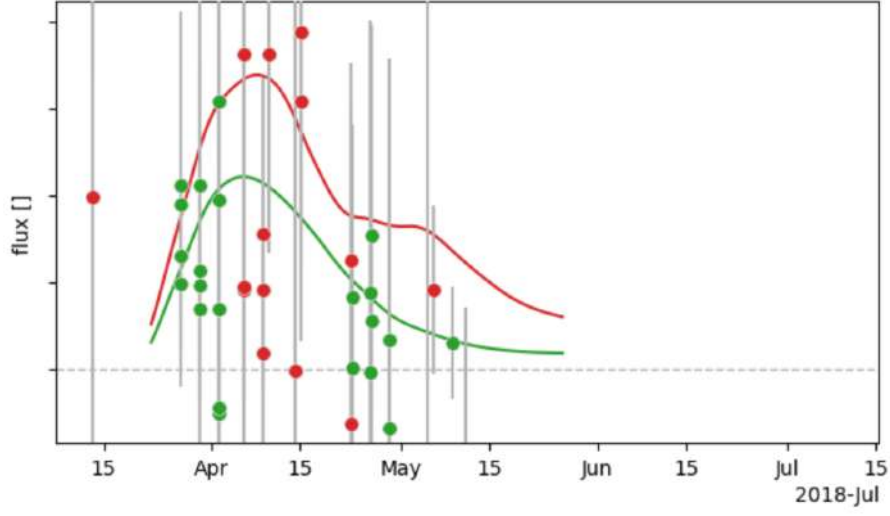


Figure 4: Light curve of non-type Ia-supernova "ZTF18aaapivw" at estimated redshift $z = 0.023$. The photons in the two g and r band filters are shown. The points represent the photons data while the lines represent the best physical light curve fitting.

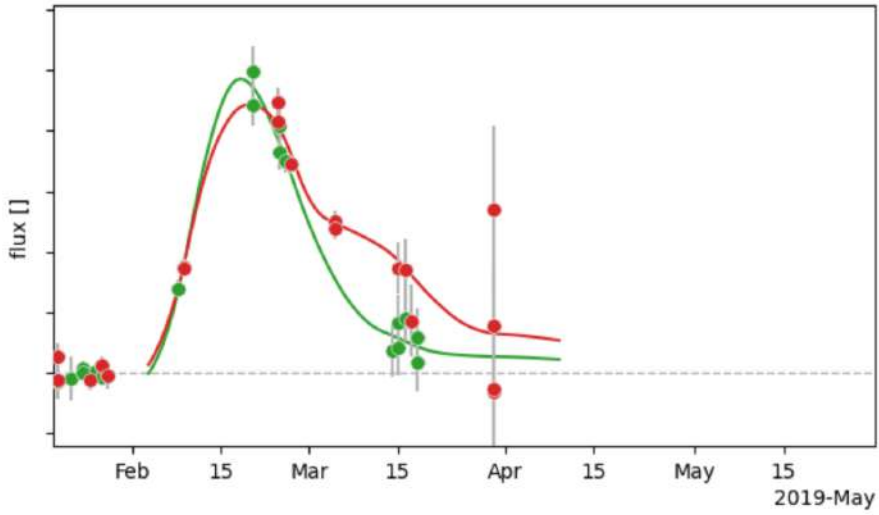


Figure 5: Light curve of non-type Ia-supernova "ZTF19aahsclk" at estimated redshift $z = 0.059$. The photons in the two g and r band filters are shown. The points represent the photons data while the lines represent the best physical light curve fitting.

of the predictions made by the algorithm from the ZTF sample. Overall, we obtain the first positive indications on the accuracy of the classification predictions from the ParSNIP algorithm on type Ia supernovae. The results show that the algorithm may well predict the classification of type Ia supernovae from ZTF. However, the predictions of non-Ia supernovae were not accurate, yet the reason remains relatively unclear. We must determine whether this problem arises because of bad classification after the ZTF measurements or if the algorithm simply poorly predicts all non-Ia supernovae from the sample. It is relevant to restrict further the studies on high-quality samples (with sufficient amounts of data flux points) on ZTF. It is also necessary to redo the work after pre-training the ParSNIP algorithm on ZTF. This could shed light on the wrong classification of non-Ia supernovae. Those two steps are needed to reach a significant conclusion on whether the adaptation of this algorithm to the ZTF sample is good enough.

3 Conclusions and Outlook

In this study, we used the ParSNIP algorithm, a recently developed neural networks algorithm, to generate predictions of the classification of supernovae objects registered from the ZTF dataset. Since the algorithm was adapted to the PS1 dataset, we explained how to adjust the ZTF datasets so that the algorithm can read the data and generate predictions. After revealing this procedure, we compared the results we obtained between the projections on the PS1 dataset and the ZTF dataset, in both cases, after having pre-trained the ParSNIP model to PS1. We found that the classification confirmed well in 79% of the cases the type of the Ia supernovae coming from the ZTF sub-dataset. However, in 55 % of the cases, the algorithm wrongly predicted the type of non-Ia supernovae, attributing their classification to Ia. Those findings suggest that even though the algorithm's accuracy in predicting the supernovae to be Ia is relatively high, the accuracy could be better when it comes to the prediction of non-Ia supernovae. Conducting this analysis after pre-training the ParSNIP model on the whole ZTF dataset is the crucial next step to determine whether this algorithm can be confidently adapted to further ZTF samples. As mentioned, it is also essential to re-adapt a few lines of the algorithms to make the conversion adaptable to any supernova and not remain in the Ia, non-Ia binary classification.

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